Lakshya Katewa

Junior Undergraduate | Department of Economic Sciences

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Lakshya-Katewa | Lakshya Katewa

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.S-Economic Sciences	Indian Institute of Technology Kanpur	7.1/10
2022	Class XII (CBSE)	Dundlod Public School,Jhunjhunu	98.8%
2020	Class X (CBSE)	Dundlod Public School,Jhunjhunu	96.6%

Work Experience

Backend Developer | `EleWin` *Apr'26 – Jul'26*

- Developed **REST APIs** using **Node.js**, **Express.js**, and **MongoDB** for aptitude tests, question banks, and assessments
- Architected backend services for **test creation**, **question filtering**, **answer submission**, scoring, and result management
- Reduced repeated database queries using **Redis caching** for frequently requested aptitude questions and test configurations
- Implemented **pagination** and **aggregation pipelines** for assessment history, user performance, and category-wise results
- Added **input validation**, **centralized error handling**, and reusable services to improve reliability and maintainability

Mobile App Developer|Freelancer | **ChinarAgro** *(Apr'25 - May'25)*

- Developed cross-platform **Flutter(Dart)** apps with reusable widgets, navigation, state management, and form validation
- Integrated **Firebase Authentication and Cloud Firestore** for profiles, products, orders, inventory, and real-time updates
- Integrated **Razorpay Flutter SDK** for payment initiation, success and failure handling, and automated order confirmation
- Implemented **Geolocator and Google Maps** for coordinates, distance calculation, delivery prioritization, and navigation
- Built an **Admin App with Firestore CRUD** for product pricing, inventory, order processing, and delivery-task assignment

Key Projects

In-Memory Key-Value Store | Self Project | *(Aug'26)*

- Built an in-memory key-value store using **unordered_map** and **unordered_set**, supporting **O(1)** average-time operations
- Designed **Command Parser, Storage Engine, Expiry Manager,** and **Database** components using **OOP** principles
- Implemented key expiration using **C++ chrono** and file persistence for **state restoration** across application restarts

Inventory & Order Management System | Self Project | *(Aug'26)*

- Built **inventory and order management system** using **C++** and **MySQL** for product, stock, and order management
- Designed **Product, Inventory, Order, OrderItem, and User** components using **OOP** principles and modular architecture
- Implemented **CRUD operations** and **SQL workflows** for product management, inventory tracking, and order processing
- Applied **database transactions** to maintain **inventory consistency** during order placement and stock update operations

Inflation Forecasting in India using Machine Learning | Course Project *(Jan'26-Apr'26)*

- Developed a **time-series forecasting pipeline** for India's CPI using quarterly CPI momentum and macroeconomic variables
- Engineered **lag-based features** and implemented **rolling-sample forecasting** for 1- and 4-quarter-ahead predictions
- Compared **ANN and RNN-LSTM** models with **Linear Regression, ARIMA, and SARIMA** forecasting baselines
- Achieved **0.58 RMSE** with **univariate ANN** for 1-quarter forecasting, versus **LR(0.66), SARIMA(0.79), ARIMA(0.80)**

Credit Card Fraud Detection | Self Project | *(Aug'26)*

- Built a **leakage-safe** fraud pipeline on **284K+** transactions using **stratified cross-validation** and imbalance handling
- Compared Logistic Regression, Decision Tree, KNN, and **Random Forest** using **PR-AUC**, ROC-AUC, precision, recall, F1
- Tuned **thresholds** on validation data to optimize the precision-recall tradeoff, achieving **87**% **precision** and **74**% **recall**
- Analyzed confusion matrices and **feature importance** to quantify false-positive/false-negative tradeoffs and model behavior

Technical Skills

Programming Languages	C++, Python, JavaScript, SQL, Dart
Backend & Databases	Node.js, Express.js, MongoDB, Redis, BullMQ, Firebase, Cloud Firestore, REST APIs
AI/ML	Scikit-learn, TensorFlow/Keras, Pandas, NumPy, Matplotlib, Seaborn, Statsmodels
Frontend & Mobile	React, Next.js, Flutter
Developer Tools	Git, GitHub, Docker, VS Code, Firebase Storage, Firebase Auth, Razorpay, Google Maps
Core Concepts	Data Structures & Algorithms, OOP, REST API Design, Authentication & Authorization, Caching, Asynchronous Processing, Message Queues, System Design

Relevant Courses

Data Structures & Algorithms	AI, ML, DL & Its Applications	Applied Probability and Statistics

Positions of Responsibility

Secretary | Career Development Wing, Academics and Career Council, IIT Kanpur *May'24 - Apr'25*

- Collaborated with **GeekforGeeks** for curated DSA courses worth **INR 6 L+** purchased by **800+ students** campus-wide
- Organized **8 Profile Introductory Sessions** for Intern Season with **40+ expert Speakers** attended by **900+ students**
- Managed a **3-day Career Connect** with **INR 3.4 L+ budget, 20+ speakers, 10+ sessions ,2000 + total footfall**
- Empowered **1200+ students** to significantly elevate their overall preparation level for the **2024-25 Internship Drive**

Scholastic Achievements

- Awarded **Certificate of Merit** in Science by **CBSE** board in 2020, only given to the top **0.1**% of performers in the subject
- Solved **700+ DSA problems** across **LeetCode** and **NxtWave**, strengthening problem-solving and algorithmic skills